

**UNIVERSITY OF GUYANA
FACULTY OF NATURAL SCIENCES
DEPARTMENT OF COMPUTER SCIENCE**

Course Number: CSE 2201
Course Name: **Internet Computing I**
Course Credit: 4

Description: This course is a second year second, semester course intended for students pursuing the two year full time Associate Degree in Computer Science or Information Technology program. This course introduces students to the Internet, World Wide Web and related technologies. It focuses on introducing client-side and server-side technologies, concepts and the techniques involved in using these technologies and concepts to effectively design and develop web-based interfaces.

Exemption(s): There are no exemptions for this course.

Pre-requisites:

Follow-On Courses: CSE 3101 - Internet Computing II

Learning Outcomes:

At the end of this course, students should be able to:

1. Explain the infrastructure and data transfer mechanisms of the Internet
2. Produce navigable information architectures for information-rich websites
3. Discuss contemporary issues related to web development
4. Design and evaluate usable, rich web-based interfaces
5. Design and implement a fully functional website using appropriate client-side and server side technologies

Course Content:

WEEK	TOPICS	Lecture HOURS	LAB HOURS
1	Introduction The Internet & The World Wide Web o History & evolution of the web, architecture & components, user agents, web servers, communication technologies & protocols Labs	3	2

2	Client-Side Computing & An Overview of Current Technologies - o What is Client-Side Computing, Client-side development languages overview, discussion on separation of concerns, other client-side technologies. Design considerations for client-side web development - o Overview of patterns and heuristics, web interface design heuristics Assignment # 1 Labs	3	2
3	Design considerations for client-side web development o Navigation design, information architecture and navigational maps Labs	3	2
4	Development with a Mark-up Language for structure o Introduction to syntax, conventions and usage Labs	3	2
5	Development with a Mark-up Language for structure II o Advanced constructs Test #1 Labs	2 1	2
6	Enhancing structure with a Styling Language o Styling & Structure, Introduction to syntax, conventions and usage. Assignment # 2 Labs	3	2
7	Enhancing structure with a Styling Language II - o Advanced constructs Website Evaluation - o Usability evaluation, General performance evaluation, web metrics Labs	3	2
8	Development with a Scripting Language o Introduction to syntax, conventions and usage.	2	2

	Test # 2 Labs	1	
9	Development with a Scripting Language II - o Advanced constructs, integration with other languages and technologies - o Rich web interfaces and design patterns, Special topics Assignment # 3 Labs	3	2
10	Server-Side Computing & An Overview of Current Technologies - o What is Server-Side Computing, Server-side development languages overview, discussion on separation of concerns, server-side technologies. - o Databases and web applications Labs	3	2
11	Development with a Server Side Language I o Introduction to syntax, conventions and usage Labs	3	2
12	Development with a Server Side Language II o Introduction to syntax, conventions and usage Labs	3	2
13	Development with a Server Side Language III o Advanced constructs Labs	3	2
14	Revision		
15	Final Examination		
Total	65 Hours of Contact	39	26

Method of Teaching:

Lectures 3 x 13 = 39 hrs.

Laboratories/ Tutorials 2 x 13 = 26 hrs.

Method of Assessment:

Coursework (40%)

- Tests (20%)
- Assignments (20%)

Final Examination (60%)

Recommended Reading(s)

- Deitel, H. M., Deitel, P. J., & Nieto, T. R. (2012). Internet & world wide web (5th ed.). Prentice Hall.
- Sebesta, R. W. (2015). Programming the world wide web (8th ed.). Pearson Addison Wesley.